

VARNA, Bulgaria

WMO: 155520

Lat: 43.232N

Lon: 27.825E

Elev: 230

StdP: 14.57

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.6	19.7	8.7	8.7	17.3	13.8	11.2	22.5	34.6	28.2	31.1	29.4	12.7	0	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.2	89.6	71.7	87.0	71.6	84.5	70.9	76.2	84.5	74.5	82.8	72.9	81.1	10.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.6	126.1	81.3	71.7	118.3	79.9	70.0	111.4	78.1	39.8	84.6	38.2	82.8	36.7	81.5	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	22.2	18.5	DB	9.1	95.9	4.9	4.2	5.6	98.9	2.8	101.4	0.0	103.8	-3.5	106.9	(4)
(5)			WB	8.3	79.3	4.9	1.7	4.8	80.5	2.0	81.5	-0.8	82.4	-4.3	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.1	36.1	38.8	44.7	52.2	61.4	69.6	74.1	74.2	66.1	56.4	48.0	39.0		
	DBStd	14.97	8.66	8.64	7.25	6.11	5.47	4.76	3.87	4.36	5.42	6.74	7.99	8.39		
	HDD50	1499	432	323	197	45	1	0	0	0	0	23	127	351		
	HDD65	4440	895	734	631	386	136	14	1	1	52	273	511	806		
	CDD50	3376	3	8	31	110	355	588	747	751	484	222	67	10		
	CDD65	841	0	0	0	1	25	152	282	287	86	7	1	0		
	CDH74	6912	0	0	5	24	171	1125	2441	2557	537	49	3	0		
	CDH80	2047	0	0	0	3	20	265	787	876	92	4	0	0		
	WSAvg	7.6	8.1	8.4	8.3	7.7	6.9	7.0	7.2	7.3	7.7	7.6	7.4	8.2		
	PrecAvg	21.0	1.7	1.3	1.6	1.6	1.7	1.9	1.8	1.3	2.2	2.2	1.9	1.9		
(15) Precipitation	PrecMax	34.4	4.3	3.7	4.4	4.6	4.0	7.8	4.3	4.5	6.1	8.1	6.5	6.2		
	PrecMin	10.9	0.1	0.1	0.0	0.3	0.6	0.4	0.0	0.0	0.1	0.1	0.1	0.0		
	PrecStd	5.2	1.3	0.9	0.9	0.8	0.9	1.4	1.3	1.1	1.6	1.9	1.5	1.5		
	PrecStd	5.2	1.3	0.9	0.9	0.8	0.9	1.4	1.3	1.1	1.6	1.9	1.5	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.0	66.3	73.1	77.2	82.5	89.6	93.6	94.6	86.3	79.2	73.3	63.0		
		MCWB	52.0	54.3	54.9	60.7	67.2	72.0	71.5	72.3	67.8	66.4	60.7	54.7		
	2%	DB	55.6	59.4	65.9	71.3	78.5	85.8	89.5	90.0	82.5	74.0	67.7	59.0		
		MCWB	48.7	50.7	53.3	58.0	64.8	70.8	72.0	71.8	68.2	63.7	59.2	52.5		
	5%	DB	51.4	55.1	60.8	66.4	75.3	82.7	86.4	87.6	79.7	70.4	62.9	55.1		
		MCWB	46.5	48.5	50.7	55.9	63.9	69.8	71.8	72.0	67.9	62.3	57.4	50.5		
	10%	DB	47.9	50.3	55.8	62.6	72.3	80.5	84.3	84.6	77.0	67.6	59.3	51.6		
		MCWB	45.1	45.5	48.1	54.2	62.6	69.1	71.1	71.4	66.7	60.9	55.2	48.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	55.2	57.0	62.2	70.0	76.0	78.0	79.3	74.0	69.0	62.5	56.7		
		MCDB	58.8	62.8	69.1	73.6	77.8	84.7	86.4	86.6	80.6	75.7	69.1	61.8		
	2%	WB	49.8	51.6	53.9	59.2	67.6	73.6	75.9	76.8	71.6	65.8	60.2	53.7		
		MCDB	54.4	58.0	64.1	68.7	75.3	81.6	84.4	85.0	78.3	71.4	65.6	57.7		
	5%	WB	47.3	48.8	51.3	57.0	65.7	71.9	74.3	74.8	69.8	63.6	58.2	51.1		
		MCDB	50.3	54.0	59.1	64.8	72.9	79.9	82.9	83.0	76.2	68.7	62.8	53.8		
	10%	WB	44.9	46.1	49.2	55.0	63.8	70.2	72.7	73.1	68.2	61.5	55.8	48.6		
		MCDB	47.1	49.4	55.1	61.4	70.5	78.1	81.2	81.4	74.6	66.1	59.1	51.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	15.7	17.7	18.5	19.3	19.6	20.5	21.2	19.4	17.4	15.1	14.5		
		MCDBR	20.5	23.8	25.7	24.6	23.5	23.0	23.3	23.9	23.0	22.7	20.9	19.4		
		MCWBR	14.0	15.2	14.6	13.5	13.3	12.6	12.5	12.8	12.8	13.9	12.9	13.4		
		MCDBR	18.0	22.2	23.5	22.4	21.4	21.1	21.3	22.0	21.3	21.0	18.6	16.6		
	5% WB	MCWBR	13.1	14.6	14.0	13.2	12.9	12.7	12.4	12.9	12.8	13.8	12.0	12.5		
(40) Clear-Sky Solar Irradiance	taub		0.351	0.381	0.415	0.460	0.433	0.436	0.445	0.455	0.432	0.398	0.373	0.353		
	taud		2.327	2.247	2.179	2.064	2.225	2.254	2.243	2.210	2.241	2.312	2.322	2.329		
	Ebn at Noon		241	253	259	256	267	266	262	255	251	244	230	228		
	Edh at Noon		27	34	41	50	44	43	43	43	38	32	27	25		
(44) All-Sky Solar Radiation	RadAvg		454	685	1064	1515	1910	2097	2112	1893	1362	858	515	395		
	RadStd		61	86	105	158	123	144	125	94	115	87	69	55		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-184	+169	+90
(47)	Regional (28 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	-205	+155	+94

Nomenclature: See separate page